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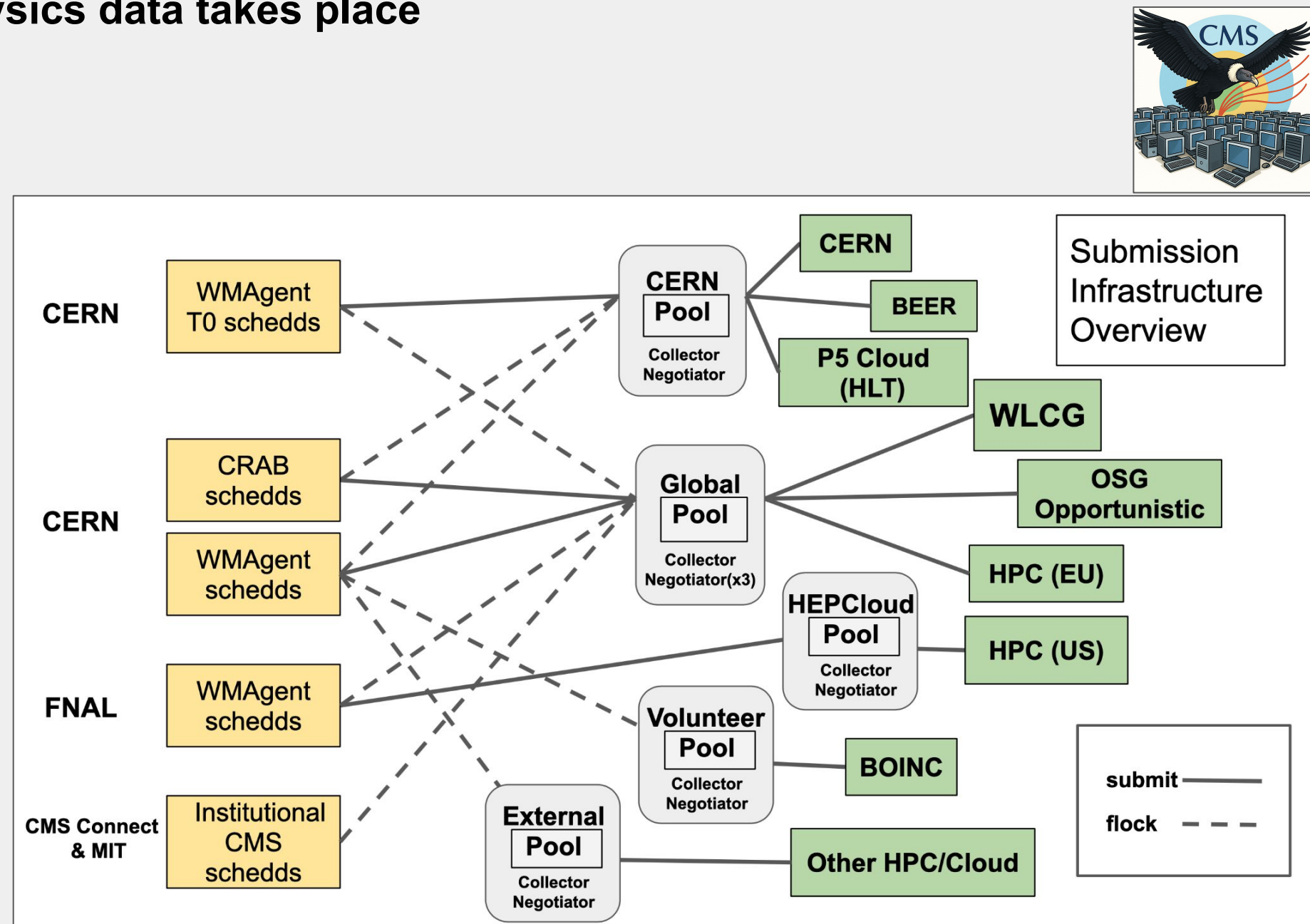
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The CMS Submission Infrastructure

The CMS Submission Infrastructure (SI) team is in **charge** of operating a set of federated HTCondor pools aggregating **resources from 70 Grid sites, plus non-Grid resources, where reconstruction, simulation, and analysis of CMS physics data takes place**

The challenges:

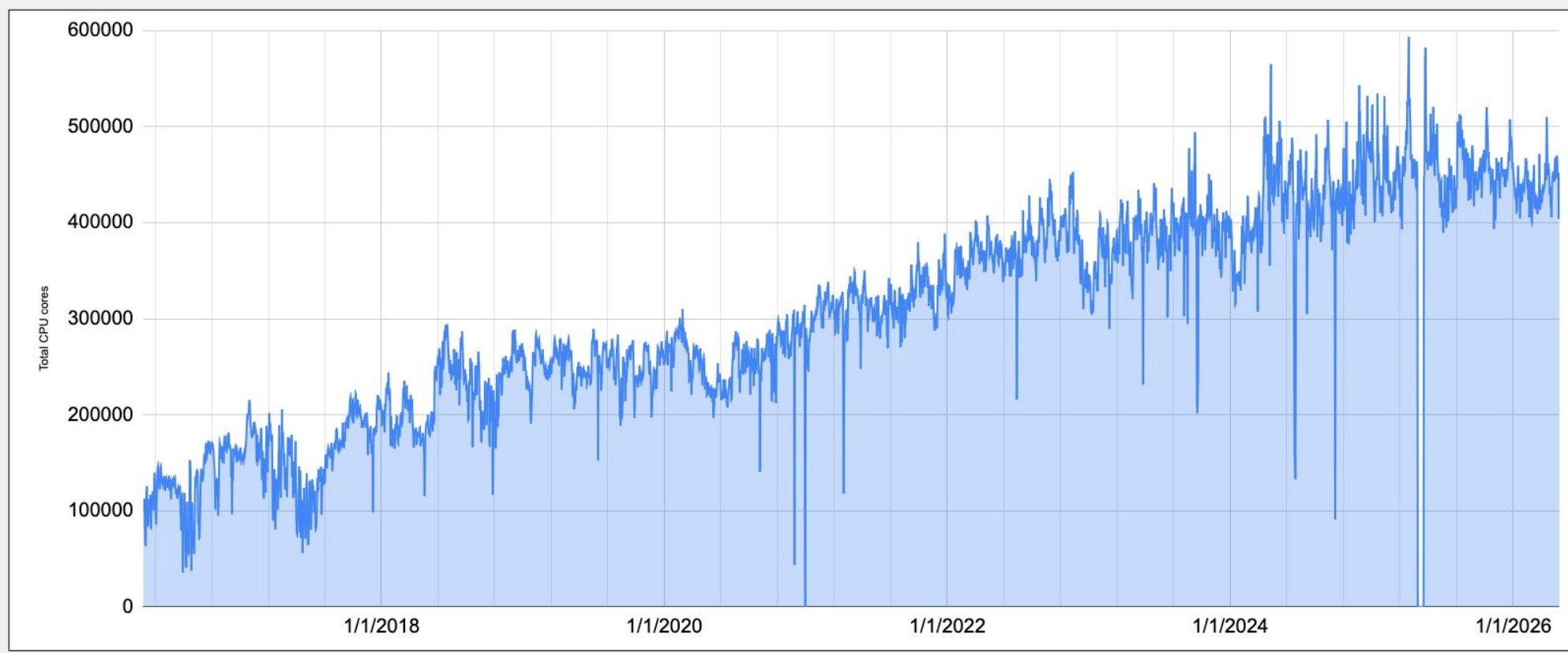
- Operate our infrastructure managing an ever **growing** collection of computing resources
- Connecting new and more **diverse resource types** (including non-x86 CPU architectures and GPUs) and **resource providers** (WLCG and OSG, HPC, Cloud, volunteer)
- Use all of our resources efficiently**, maximizing data processing throughput
- Enforce **task priorities** according to CMS research programme



Scalability in the CMS SI

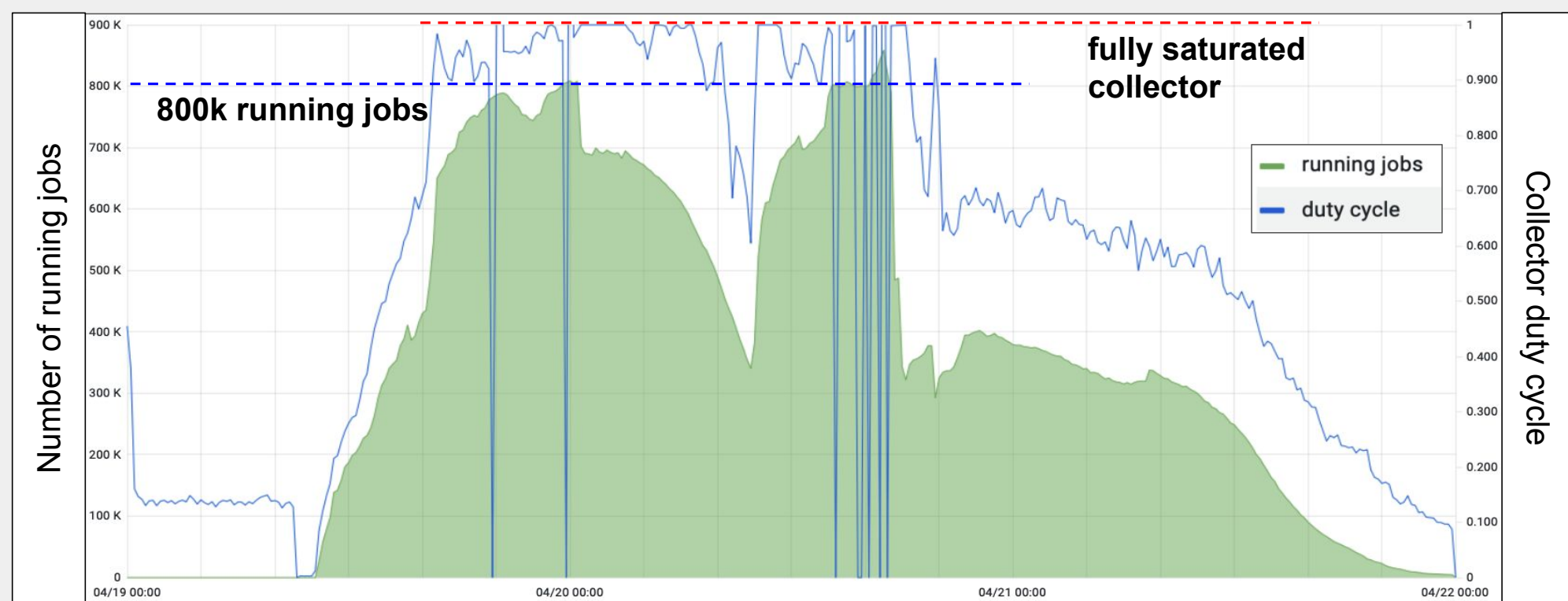
- The system is designed and deployed to operate away from scalability limitations, which is critical to perform in a dynamic environment, with evolving **resource demands** by CMS, and **resource availability** in the WLCG.
- The SI team works to **proactively find those limits**, evolving the infrastructure to push them further away:
 - Total computing power our HTCondor pools can harness and use efficiently
 - Collector capacity to process the stream of slot updates and keep resource status fresh
 - Negotiator matchmaking cycle time under control
 - Total number of workflows that can be managed, and jobs executed simultaneously with our pool of **scheduler** nodes

Total resources in CPU cores allocated to CMS over the past years (daily averages)



Pushing the limits of the CMS Global Pool

- Since the start of the Run 2, we have been continuously detecting and solving bottlenecks to our Global Pool, with the support of the HTCondor and GlideinWMS developers teams over the years
- Multiple **customized settings** directed at **avoiding the saturation of the top collector**:
 - Using a **CCB** running on a separate host to the CM, with enlarged pool of connection sockets
 - Use of multiple **negotiator** daemons running in multithreaded mode
 - Hierarchy of **secondary collectors** connected to the main top collector process
 - Optimized **slot update conditions** (filter on update triggers, use UDP instead of TCP, enlarged UDP buffer)
 - Classify and redirect non-high prio queries** from main to backup secondary collector (HA infrastructure at FNAL)



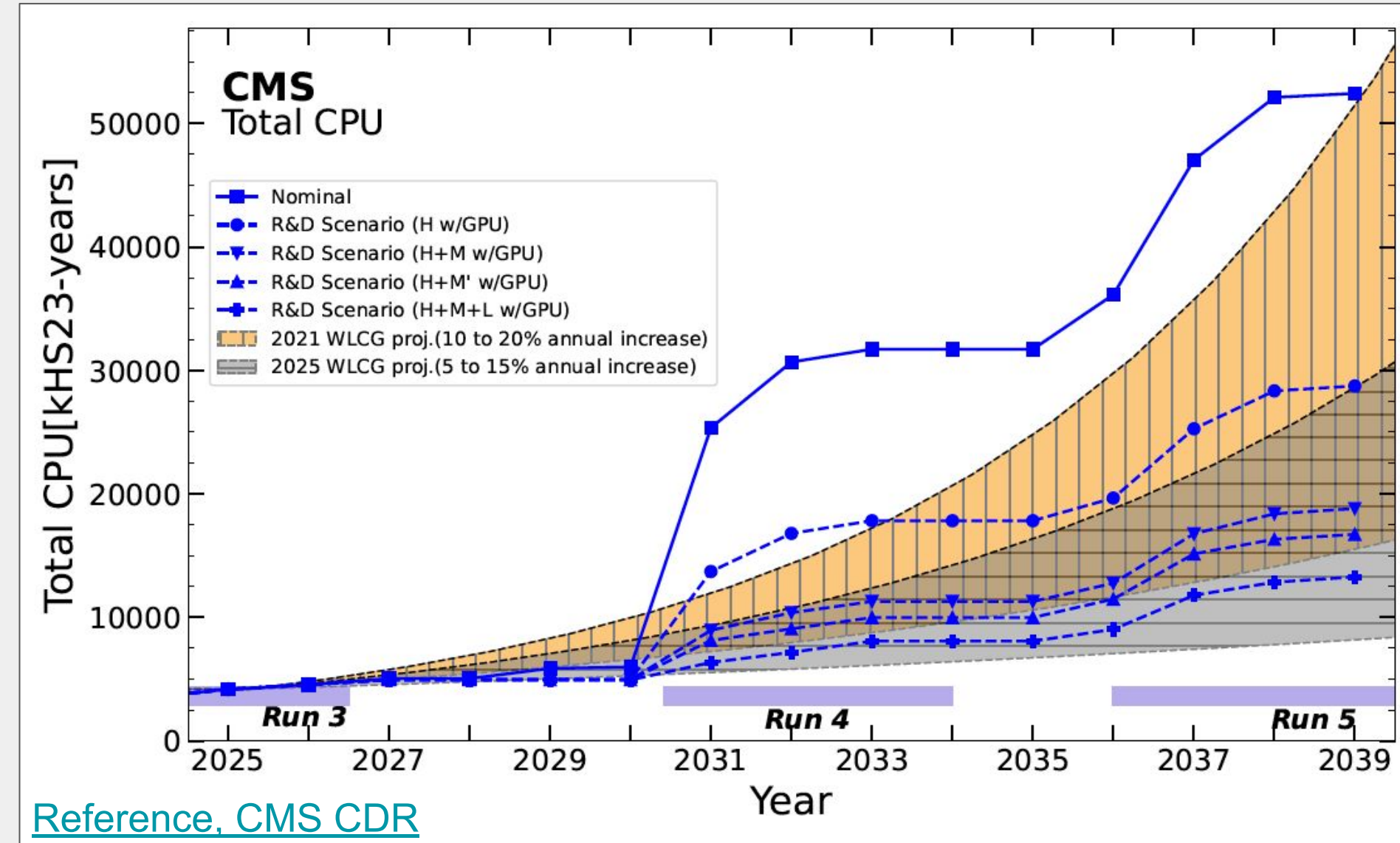
Latest collector scalability tests in 2023: 800,000 simultaneous running jobs

- See [talk](#) at 2023 HTCondor workshop
 - See CHEP 2023 proceedings [note](#)
- Previous tests references:
<https://doi.org/10.1051/epjconf/201921403002> (2018)
<https://doi.org/10.1051/epjconf/202024503016> (2019)
<https://doi.org/10.1051/epjconf/202125102055> (2021)

Increased scalability for HL-LHC

Increased resource scale and heterogeneity in the future HL-LHC phase

- Overall resource growth driven by the unprecedented data stream expected for the HL-LHC phase
- More diverse resources providers, as HPCs are expected to play a major role in the future scientific computing infrastructure
- A growing fraction of the compute power in the form of non x86_64 resources



The SI vision for HL-LHC:

- Build from the current system that has served CMS effectively for more than a decade
- Expand and adapt to more and new resource types, diverse allocation policies and tools, etc
- Preserve the demonstrated stability, flexibility and efficiency at **larger scales**

Scalability considerations from HPC resources

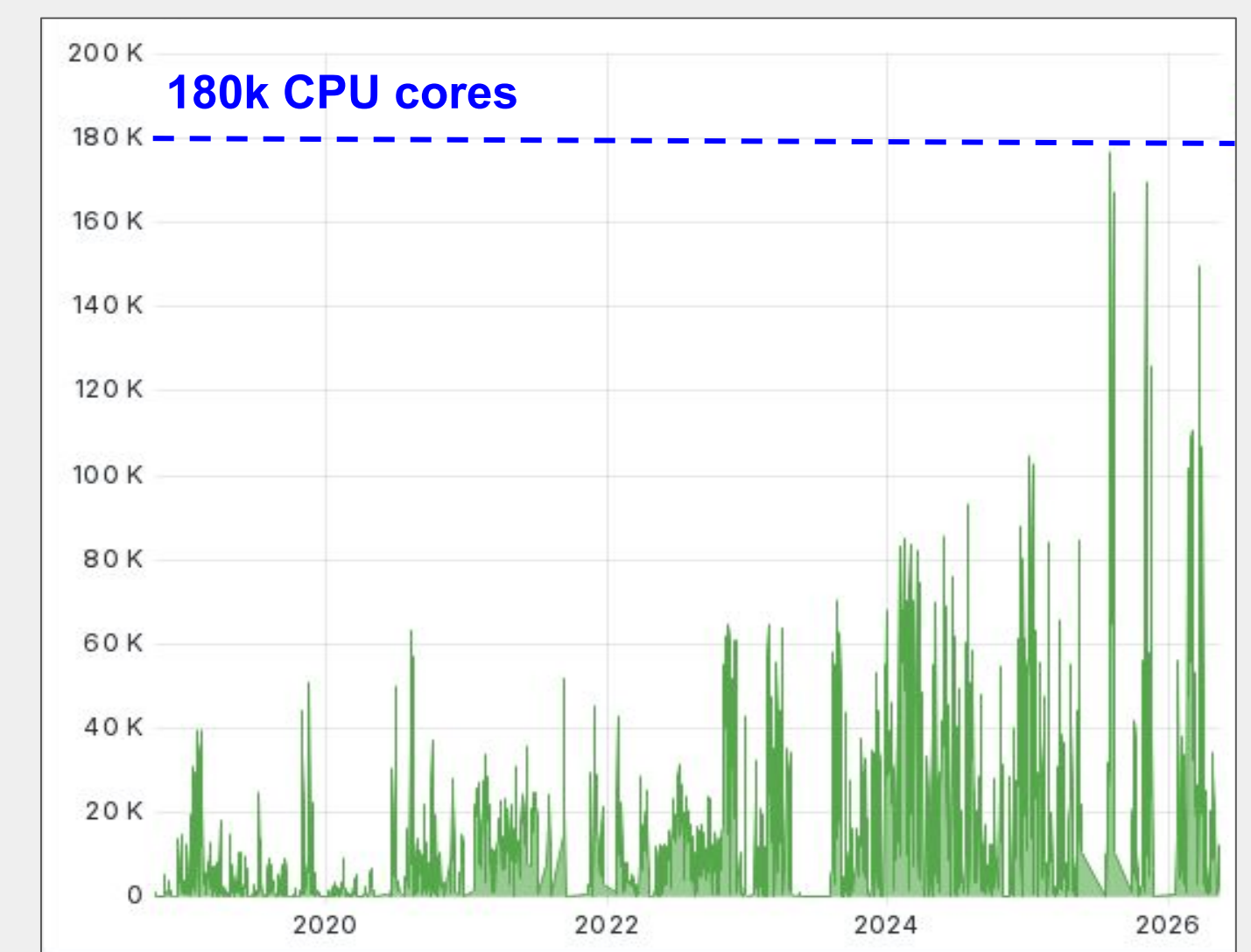
Scalability associated to HPCs resources exploitation:

- Continuous growth of baseline WM and SI capacity** to manage and supply workloads to fill increasingly larger pools of resources (*steady growth*)
- HPC vs WLCG provisioning**: WM and SI capacity should be dimensioned to profit from HPC **resource peaks** (*bursts*)

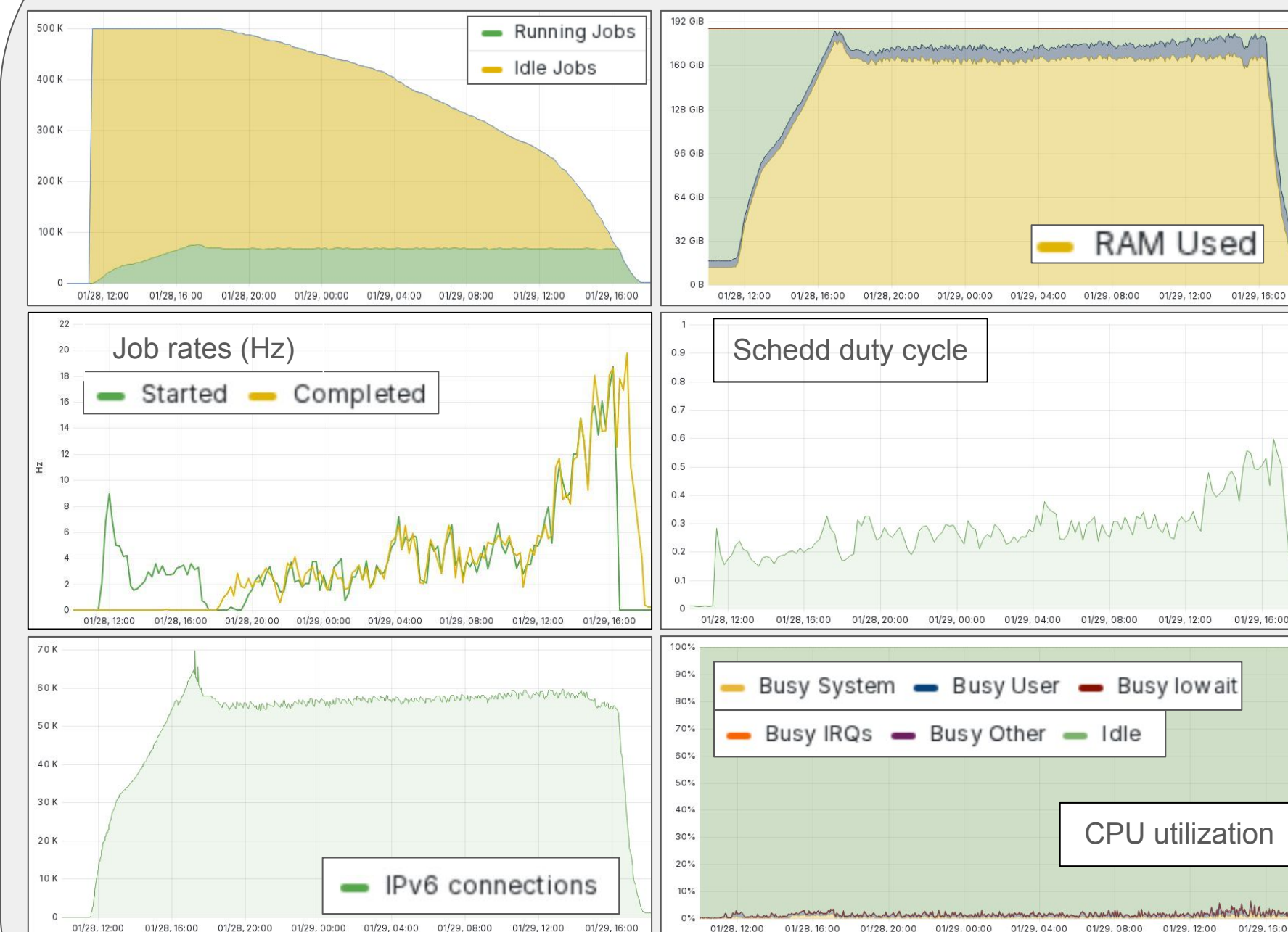
CMS SI and WM systems should ensure the effective use of CMS compute capacity enlarged with HPCs:

- Workload queue length
- Number and capacity of scheduler agents

Growing bursts of CPU allocated for CMS use at NERSC over the last years



Scalability tests of the scheduler nodes (I)



Test schedd: physical node with 196 GB memory, 2x16 cores (64 threads), 1,920 GB disk

Test setup:

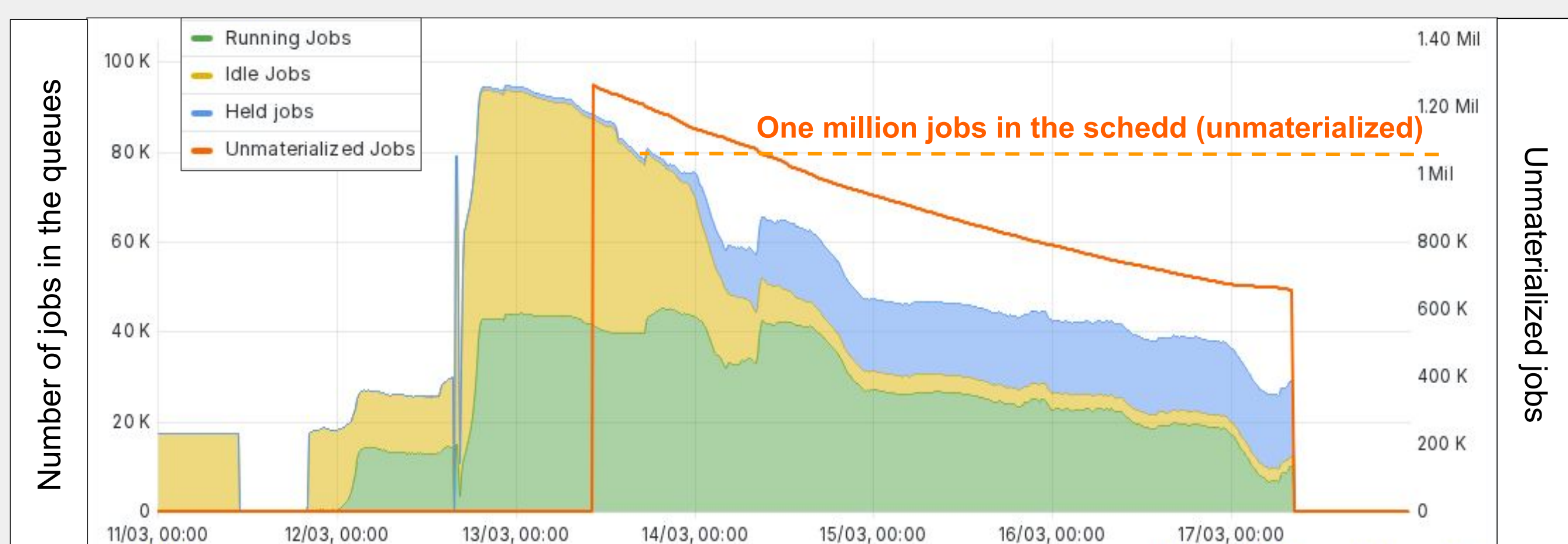
- 500k jobs** submitted with a mix of resource requests and execution time (gaussians with $\mu=1h, 4h$ and $8h$)
- Achieved 70k maximum running jobs**

Conclusions from the tests:

- Memory scales with number of running jobs, at **2MB mem/job** ratio
- Schedd **duty cycle** correlates to job starting/completion rate. Still, below 0.6 even for highest tested rates (completion at 20 Hz)
- No issue in terms of **CPU load**
- Network**, not a problem, enough IPv6 TCP sockets

Scalability tests of the scheduler nodes (II)

Late materialization of jobs in the scheduler provides additional **flexibility** to the Workload Management model, as the pool of scheduler nodes can hold **millions of jobs** with minimal resource consumption, allowing the WM system to push large workloads into the HTCondor queues, **removing unnecessary throttling**



Conclusions and Future work

- The CMS SI team, in collaboration with the HTCondor and GlideinWMS developers, **periodically evaluates the scalability** of our infrastructure to **anticipate** and **remedy** future scaling problems and stay off the bleeding edge of limitations
- In previous scalability tests we concentrated on the central manager of the HTCondor pools (**collector and negotiator**), which in latest test (2023) demonstrated the capacity to support up to **0.8M simultaneous running jobs**
- With the current tests, we have explored the scalability of the scheduler nodes of the pool**, as these agents will be responsible to fill growing pools of resources, even in burst mode, in future scenarios
- The LHC program extends well into the future, so we need to **preserve the scalability of the CMS SI for HL-LHC scales**, as required by CMS needs, while **maintaining stability and efficiency**

